

Draft - expert review pending

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Draft translation - not yet human-reviewed / مسوده ترجمه

Topic 3.1 - Learning as behaviour: Pavlov and Skinner

A real classroom situation

Two things happen in Miss Farzana's Class 5 in a government school in Tando Allahyar, and both happen every day.

First: the pupils can hear the head teacher's footsteps on the corridor before he appears. By the time he opens the door, the room is already silent. Nobody taught them to fall silent at footsteps; the sound simply arrived, many times, shortly before a scolding.

Second: Miss Farzana keeps a star chart. A pupil who completes the day's work gets a star beside their name, and five stars mean a turn watering the class plants, the most wanted job in the room. Work completion has risen since the chart appeared.

Both events are learning, and neither involved a lesson. The staff room has one word for the first ("discipline") and another for the second ("incentives"), but they are two mechanisms of the same family - and the family has a name.

before conditioning



after conditioning



bell: neutral stimulus becomes conditioned stimulus

food: unconditioned stimulus

salivation: response now triggered by the bell

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Explanation

Learning as behaviour: the behaviourist view

Behaviorism is the view of learning that focuses on **changes in observable behaviour** - changes in what people say or do (Seifert & Sutton, 2009). It began as a deliberate decision by psychologists: thinking cannot be seen, so a science of learning should start from what can. Whatever its limits, every teacher uses this perspective daily. When you check whether pupils *can do* the sum, rather than whether they can describe how to do it, you are measuring learning behaviourally.

The behaviourist family has two great mechanisms. The first explains behaviour that pupils **cannot help** - the falling silent at footsteps. The second explains behaviour pupils **choose** - the work done for a star. Confusing the two is the commonest error in classroom management, because the two call for different teacher moves.

Pavlov: classical conditioning

Classical conditioning is learning by **association**: a neutral signal comes to trigger a response because it repeatedly arrived just before something that already triggered it. Ivan Pavlov, a Russian physiologist, demonstrated it with dogs: a bell rang before food; after repetitions, the bell alone made the dogs salivate (Seifert & Sutton, 2009).

The mechanism needs only four names. The **unconditioned stimulus** (food) already triggers the **response** (salivation) without learning. The **neutral stimulus** (the bell) triggers nothing - until it repeatedly precedes the food, at which point it becomes a **conditioned stimulus** that triggers the response on its own.

Now read the Class 5 corridor scene with those four names. The scolding is the unconditioned stimulus; fear and stillness are the response; the footsteps are the neutral stimulus that became conditioned. The pupils did not decide to fall silent any more than Pavlov's dogs decided to salivate. That is why telling them to relax is useless: nobody can un-hear an association by being told to.

Classical conditioning also fades when the signal stops predicting: if the footsteps repeatedly led to nothing, the silence would wear off. Psychologists call this **extinction**, and it matters to a teacher in one specific way - associations the school creates by accident (fear of the head teacher, dread of a subject) fade only when the pairing stops, not when anyone announces it should.

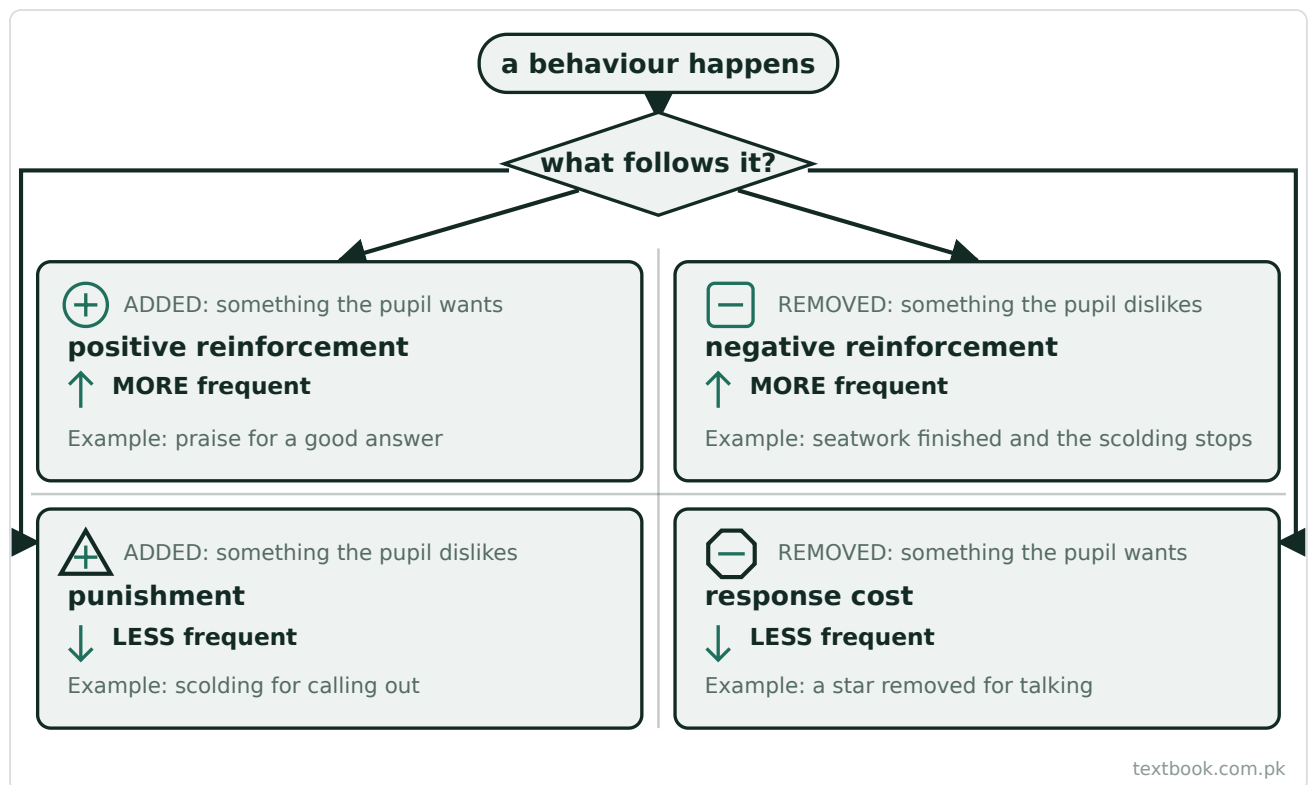
The misconception to clear now. Trainees often say conditioning is "for dogs, not children". But the corridor scene is conditioning in a Sindh government school, and so is the pupil who feels sick at the smell of the medicine she took during a hard school month. The honest objection is different: conditioning explains only part of human learning - the automatic part. It says nothing about understanding, and Topic 3.2 is where the course answers it.

Skinner: operant conditioning

Operant conditioning is learning from **consequences**: behaviour that is followed by good outcomes happens more often; behaviour followed by bad outcomes happens less often. B. F. Skinner, a Harvard psychologist, mapped the process in painstaking detail with laboratory rats and then argued, in book after

book, that the same laws run through human behaviour (Seifert & Sutton, 2009).

The key word is **consequence**, and it must be read precisely. A consequence is what actually follows the behaviour - not what the teacher intends, and not what the teacher says will follow. Miss Farzana's star is a consequence because it actually arrives. A promised prize that never materialises is not a consequence; it is a broken signal, and broken signals teach pupils to ignore the teacher's words.



Reinforcement and punishment in the classroom

The four consequence types are worth naming exactly once and then using forever (Seifert & Sutton, 2009).

Reinforcement always means one thing: the behaviour becomes **more** frequent. It comes in two forms. **Positive reinforcement** adds something the pupil wants - praise, a star, a turn at the plants. **Negative reinforcement** removes something the pupil dislikes - the pupil who finishes seatwork and thereby stops the teacher's scolding has been negatively reinforced for

finishing. Both forms strengthen behaviour; only the direction of the arrow differs.

Punishment always means the behaviour becomes **less** frequent. It too comes in two forms: adding something disliked (the scolding) or removing something wanted (the star taken away, called response cost).

The classic error is reading "negative" as "bad". Negative reinforcement is not punishment; it is one of the two strengtheners. The pupil who learns to bring a notebook because forgetting it brings a scolding, and who then stops being scolded once the habit is in place, is being shaped by both mechanisms in sequence.

Three cautions belong in any classroom use.

Reinforcement teaches what is rewarded, not what is meant. The star chart rewards completed work, and completed work rose. If neat copying completes work faster than thinking, copying is what the chart teaches. The chart is not wrong; it is precise, and its precision must be aimed.

Punishment suppresses; it does not teach. A scolded behaviour goes underground, not away, and it carries a price: the pupil learns to avoid the scolding teacher, the subject, and the risk of trying. Punishment has a place, but the place is narrow, and the replacement behaviour must be taught and reinforced beside it.

Attention is a reinforcer, and the weakest pupils get it for the wrong things. A pupil who is ignored when working and attended to when disrupting is on a schedule that trains disruption. The remedy is not less attention but attention aimed - greeting the pupil at the door, noticing the work, and letting the disruption meet a calm, small response.

consequence	what it is	what it makes more or less likely	what it risks
praise for a good answer	something wanted is added	good answers more frequent	praise inflation, dependence on the teacher
star chart for completed work	something wanted is added	completed work more frequent	copying rewarded if completion is all that is checked
scolding stops when seatwork is done	something disliked is removed	finishing work more frequent	pupils only work when scolded
scolding for calling out	something disliked is added	calling out less frequent	fear, avoidance of the subject, underground behaviour
a star removed for talking	something wanted is removed	talking less frequent	the chart becomes a threat rather than a prize

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Activity: Audit the day's consequences

In pairs, about 25 minutes.

List eight consequences that actually followed pupil behaviour in a school day you know well - your own class, your practicum, or your own schooling. For each, name the mechanism (classical conditioning, positive or negative reinforcement, punishment, response cost) and say what behaviour it is training, whether or not anyone intended that.

Then find the **one consequence you would change first**, and say exactly what you would do instead and what behaviour your change would train.

Check your understanding

1. Define behaviourism in one sentence, and say what it deliberately leaves out. (*Understand*)

2. Name the four elements of the corridor scene in conditioning terms: footsteps, scolding, falling silent, and the repeated pairing. (*Understand*)
3. A pupil feels sick at the smell of the school medicines. Explain the chain that produced this, and name the one thing that will make it fade. (*Apply*)
4. Explain, in two sentences, why negative reinforcement is not punishment, with one classroom instance of each. (*Understand*)
5. The star chart raised work completion but copying also rose. Using the precision warning in this topic, say what the chart is actually reinforcing and how to re-aim it. (*Apply*)

Summary

Behaviourism defines learning as change in observable behaviour and offers two mechanisms: classical conditioning, learning by association, which explains the automatic responses no one decided to learn; and operant conditioning, learning from consequences, which explains the behaviour that is strengthened or weakened by what actually follows it. Reinforcement (positive and negative) makes behaviour more frequent; punishment and response cost make it less. The classroom discipline is to read consequences for what they actually train - not what they are meant to teach - and to remember that attention itself is a reinforcer aimed by the teacher's choices.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can define behaviourism and name the two mechanisms with a classroom example of each.
- ☐ I can label a conditioning chain with its four elements, including from a real school scene.
- ☐ I can state the reinforcement-punishment distinction precisely, including why negative reinforcement strengthens.
- ☐ I can audit a real classroom consequence for what it actually trains, intended or not.

Try this at your practicum school

Choose one pupil everyone labels "difficult". For one day, note what behaviour of theirs actually gets attended to, and what gets ignored. Do not intervene. At the end of the day, name the schedule you have observed: what is that pupil being reinforced for, and by whom?

Summative task

Write **250-300 words** redesigning Miss Farzana's system.

The star chart raised completed work but you have noticed pupils rushing and copying. Using this topic's precision rule, analyse what the chart actually reinforces, then redesign it: what exactly will earn a star, what will be checked before it is given, and what behaviour the redesigned chart trains. Name any risk your redesign carries.

Mini-rubric

Criterion	What a strong answer shows
Mechanism accuracy	Reinforcement is used precisely; the analysis names what is actually rewarded, not what was intended.
Redesign precision	The new rule states the exact behaviour, the exact check, and the trained outcome. (<i>Analyze</i>)
Risk awareness	At least one honest cost of the redesign is named.
Plain, accurate writing	Terms used correctly; no padding; within the word count.

Further reading

- Seifert, K., & Sutton, R. (2009). *Educational Psychology* (2nd ed.), Chapter 2. University of Manitoba. CC BY.

<https://open.umn.edu/opentextbooks/textbooks/educational-psychology>

- Vosniadou, S. (2001). *How children learn*. Educational Practices Series No. 7. International Academy of Education / UNESCO IBE. <https://eric.ed.gov/?id=ED455955>

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